2 A bungee jumper of mass 60 kg is attached to an elastic rope which starts to stretch after a short time of free fall. The gravitational potential energy of the bungee jumper is 0 J when she has fallen through 40 m to the lowest point.

(a)

	gravitational potential energy / kJ	elastic potential energy / kJ	kinetic energy / kJ
top		0	0
half-way		2.6	
bottom	0		

[3]

Fig. 2.1

Fill up the missing energies at the top, bottom and half-way positions in Fig 2.1. Drag forces can be considered negligible.

(b) Show that the unstretched length of the elastic rope is 10 m.

[2]

(c) Using **(b)**, determine the spring constant, k of the elastic rope.

k = ...N m^{-1} [2]

(d) Determine the extension of the elastic rope when the kinetic energy of the bungee jumper is the highest.

extension = ... m [2]

[Total: 9]